DrumFeatureDemonstrator Documentation

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October 7, 2026, for DrumFeatureDemonstrator 0.0.1

1 Introduction

DrumFeatureDemonstrator is a one-sample drum player with a step grid (1/16 notes, 8 to 32 steps). You load one sample (a drum hit, a cymbal, a reversed cymbal), choose the steps on which it is played, and change pitch, direction and loudness per step. The sample is also played from MIDI notes.

The new thing is **play to end**. The plug-in knows exactly how long the sample sounds at the chosen pitch. A step in the mode *reverse + play to end* is therefore started early, so that the sample *ends exactly at the end of that step*. Put it on the last step before a beat, and a reversed cymbal leads into the next beat without any manual nudging, at any pitch and any tempo. The plug-in reads the beat position from the DAW, so it always stays in sync.

2 Installation

The downloaded file is a zip file. It contains the plug-in (.vst3; on macOS also the AU, .component), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place and let your DAW rescan its plug-ins.

```
Windows (VST3):  C:\Program Files\Common Files\VST3
Mac (VST3):      /Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3
Mac (AU):        /Users/yourUSERNAME/Library/Audio/Plug-Ins/Components
Linux (VST3):    /home/yourUSERNAME/.vst3
```

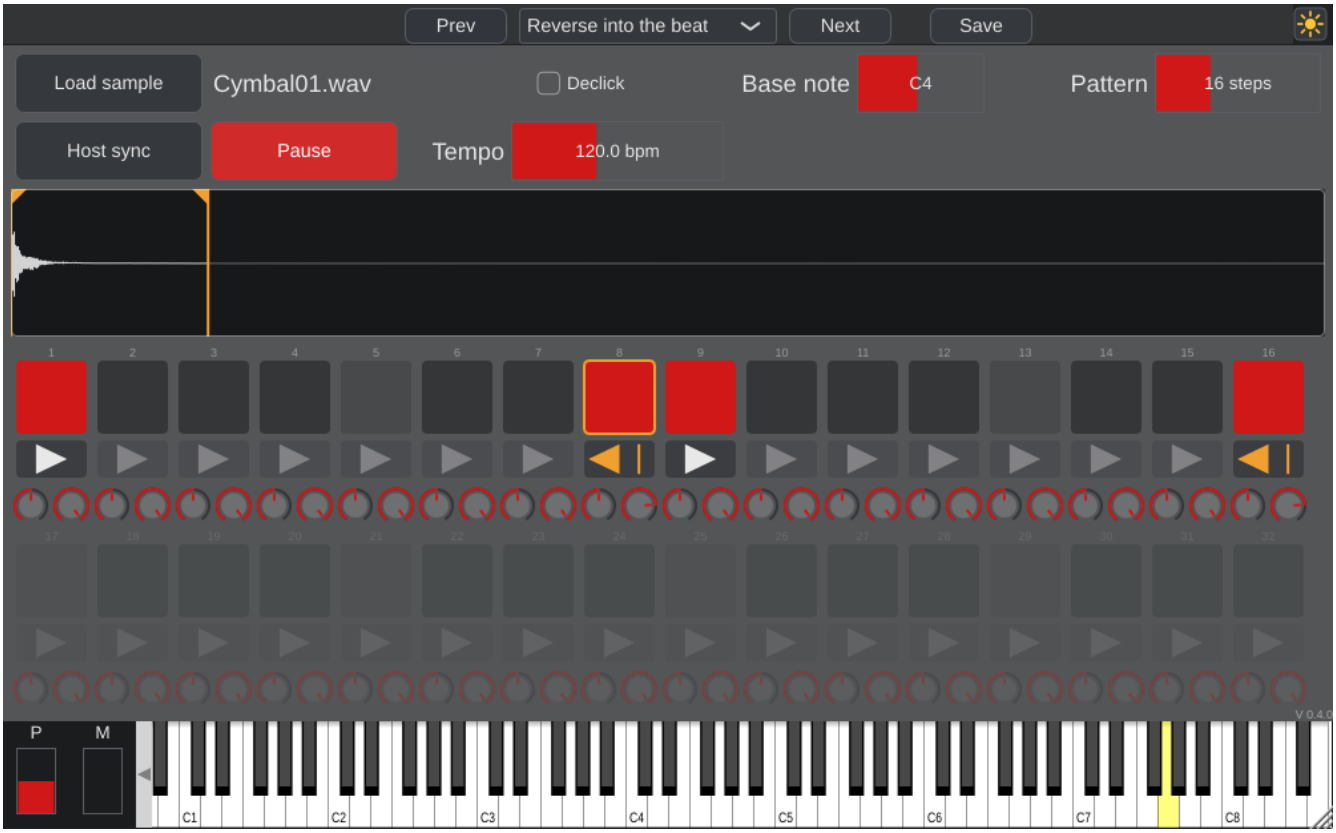
The macOS version is not signed with an Apple Developer ID. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g. `xattr -cr ~/Library/Audio/Plug-Ins/VST3/DrumFeatureDemonstrator.vst3`.

2.1 Presets and settings

The presets (plain XML files) and the settings file (`user.settings`: day/night theme) are stored in this folder – delete it to remove DrumFeatureDemonstrator completely:

```
Windows: C:\Users\yourUSERNAME\AppData\Roaming\Jade_Hochschule\DrumFeatureDemonstrator
Mac:     /Users/yourUSERNAME/Library/Audio/Presets/Jade_Hochschule/DrumFeatureDemonstrator
Linux:   /home/yourUSERNAME/.config/Jade_Hochschule/DrumFeatureDemonstrator
```

3 How to use



1. **Load sample**: choose a sample file (wav, aiff, flac, ogg), or drop a file onto the waveform. A copy is kept in the folder `Samples` next to the presets; the sample is stored with your DAW project and referred to by its name in a preset.

2. **Waveform**: drag the two orange handles to set start and end of the part that is played. With **Declick** switched on, a handle snaps to a zero crossing of the sample (no click at the edge).

3. **Base note**: the MIDI note at which the sample plays at its original speed (C4 = MIDI note 60). MIDI notes (also the keyboard at the bottom) play the sample with the pitch relative to the base note and with velocity.

4. **Pattern**: length of the pattern in steps (8 to 32). Steps beyond the length are dimmed.

5. **Host sync**: on = the plug-in follows tempo and position of the DAW (it plays while the DAW plays); off = internal beat with its own tempo and the **Play/Pause** button.

6. **Steps**: per step a button (sample on/off), below it the **mode button** (click: forward → reverse → reverse + play to end → forward), and two small knobs: left the pitch (±24 semitones relative to the base note), right the velocity (gain). The orange frame shows the step that is playing.

7. **Preset handler** and **day/night**: as usual; Save turns red when you change a control. Presets contain the pattern, the region and the name of the sample. Factory presets keep the sample you have loaded.

3.1 Play to end: what to know

- The sound ends at the *end of the step* on which the mode is set. To lead into beat 1, set it on step 16 of a pattern of 16 steps.
- If the start time has already passed (the transport starts or jumps into the run-up), the sound is *not* played and a notice appears.
- Play to end is only possible if the sample (at its pitch) is not longer than the pattern. Otherwise the mode button is outlined in red and a notice is shown: shorten the sample with the end handle, raise the pitch or lengthen the pattern.
- Tempo changes during the run-up are not followed.
- Forward + play to end is not offered; reverse the sample file beforehand if you need it.

4 Controls

All controls, their range and default value. Rest the mouse on a control to see its description as a tooltip. Every step n (1 to 32) has four parameters (`Step`*nn*`Active`, `Mode`, `Pitch`, `Velocity`); they are described once in the last rows of the table.

Control	Range	Default	Description
Base note	0 - 127	60 (C4)	MIDI note at which the sample plays at its original speed.
Start	0 - 1	0	Start of the played region, as a fraction of the sample length (waveform handle).
End	0 - 1	1	End of the played region (waveform handle).
Pattern length	8 - 32 steps	16	Number of 1/16 steps of the pattern.
Sync to host	off / on	on	On: follow the tempo and position of the DAW. Off: internal tempo and play button.
Internal tempo	40 - 240 bpm	120	Tempo of the internal beat.
Internal play	pause / play	pause	Plays or pauses the internal beat (only without host sync).
Declick	off / on	off	Handles snap to a zero crossing of the sample when moved.
Step n on	off / on	off	Plays the sample on this step.
Step n mode	0 / 1 / 2	0	0 forward, 1 reverse, 2 reverse and play to end.
Step n pitch	−24 - 24 st	0	Pitch of the step in semitones, relative to the base note.
Step n velocity	0 - 1	1	Gain of the step.

5 How it works

For a step in the mode reverse + play to end the plug-in computes the *duration* of the sample in host samples,

$$d = \frac{N}{r}, \qquad r = 2^{p/12}\,\frac{f_{\text{file}}}{f_{\text{host}}},$$

with the number of samples N of the played region, the pitch p in semitones and the two sample rates. The start time is the end of the step minus d. Because d is at most one pattern long, the start lies at most one pattern pass before the step, so the plug-in looks ahead one pass. The resampling is done by linear interpolation; pitch and duration change together.

The drum samples used for the examples and tests come from MusicRadar (SampleRadar, free and royalty-free): https://www.musicradar.com/news/tech/free-music-samples-royalty-free-loops-

6 Source code

DrumFeatureDemonstrator is open source. You find the source code here: https://github.com/JoergBitzer/DrumFeatureDemonstrator. Please report bugs there, too.

7 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is Joerg Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT) and, on macOS, the Audio Unit SDK (Apache 2.0); the binaries as a whole are distributed under the AGPLv3.

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